Problem 1

$$y_i \overset{ind}{\sim} \text{Normal}(x_i^T w, \lambda^{-1})$$
$$w \sim \text{Normal}(0, \text{diag}(\alpha_1, \cdots, \alpha_d)^{-1})$$
$$\alpha_k \overset{iid}{\sim} \text{Gamma}(a_0, b_0)$$
$$\lambda \sim \text{Gamma}(e_0, f_0)$$

using the factorization $q(w, \alpha_1, \cdots, \alpha_d, \lambda) = q(w)q(\lambda)\prod_{k=1}^{d} q(\alpha_k)$, derive the optimal form of each $q$ distribution.

the optimal form of any single factor $q_j$ is the expectation of the log-joint with respect to all other variables.

log-joint: $\ln p(y, w, \alpha, \lambda) = \ln p(y|w,\lambda) + \ln p(w|\alpha) + \Sigma \ln p(\alpha_k) + \ln p(\lambda)$

$$= \sum_{i=1}^{N} \ln p(y_i|w,\lambda) + \ln p(w|\alpha) + \Sigma \ln p(\alpha_k) + \ln p(\lambda)$$

$$\propto \sum_{i=1}^{N} \left(\ln\left[(2\pi\lambda^{-1})^{-\frac{1}{2}} \cdot \exp\left(-\frac{(y_i - x_i^T w)^2}{2\lambda^{-1}}\right)\right]\right)$$

$$\propto \sum_{i=1}^{N} \left(\ln\left[\lambda^{\frac{1}{2}} \exp\left(-\frac{(y_i - x_i^T w)^2}{2\lambda^{-1}}\right)\right]\right)$$

$$\propto \sum_{i=1}^{N} \left(\frac{1}{2}\ln\lambda - \frac{\lambda}{2}(y_i - x_i^T w)^2\right)$$

$$\propto \frac{N}{2}\ln\lambda - \frac{\lambda}{2}\sum_{i=1}^{N}(y_i - x_i^T w)^2$$

$$\propto \ln\left[(2\pi)^{-\frac{d}{2}} |\Lambda|^{\frac{1}{2}} \exp\left(-\frac{1}{2}\underbrace{w^T \text{diag}(\alpha_1, \cdots, \alpha_d) w}_{\sum_{k=1}^{d} \alpha_k w_k^2}\right)\right)$$

$\Lambda = \text{diag}(\alpha_1, \cdots, \alpha_d)$

$|\Lambda| = \prod_{k=1}^{d} \alpha_k$

$$\propto \frac{1}{2}\sum_{k=1}^{d}\ln\alpha_k - \frac{1}{2}\sum_{k=1}^{d}\alpha_k w_k^2$$

$$\sum_{k=1}^{d} \ln p(\alpha_k) \propto \ln\left(\alpha_1^{a_0-1} \cdots \alpha_d^{a_0-1} \cdot \exp(-b_0(\alpha_1 + \cdots + \alpha_d))\right)$$
$$\propto (a_0 - 1)\sum_{k=1}^{d}\ln\alpha_k - b_0\sum_{k=1}^{d}\alpha_k$$

$$\ln p(\lambda) \propto \ln(\lambda^{e_0-1}\exp(-f_0\lambda)) \propto (e_0-1)\ln\lambda - f_0\lambda$$

$\ln p(y, w, \alpha, \lambda) \propto$

$$\frac{N}{2}\ln\lambda - \frac{\lambda}{2}\sum_{i=1}^{N}(y_i - x_i^T w)^2 + \frac{1}{2}\sum_{k=1}^{d}\ln\alpha_k - \frac{1}{2}\sum_{k=1}^{d}\alpha_k w_k^2 + (a_0-1)\sum_{k=1}^{d}\ln\alpha_k - b_0\sum_{k=1}^{d}\alpha_k + (e_0-1)\ln\lambda - f_0\lambda$$

$\ln q(w)$ : focus on terms containing $w$

$$\frac{N}{2}\ln\lambda - \frac{\lambda}{2}\sum_{i=1}^{N}(y_i - x_i^T w)^2 + \frac{1}{2}\sum_{k=1}^{d}\ln\alpha_k - \frac{1}{2}\sum_{k=1}^{d}\alpha_k w_k^2 + (a_0-1)\sum_{k=1}^{d}\ln\alpha_k - b_0\sum_{k=1}^{d}\alpha_k + (e_0-1)\ln\lambda - f_0\lambda$$

$$\ln q(w) \propto E_{q(\lambda),q(\alpha)}\left[-\frac{\lambda}{2}\sum_{i=1}^{N}(y_i - x_i^T w)^2 - \frac{1}{2}\sum_{k=1}^{d}\alpha_k w_k^2\right]$$

this is a quadratic form in $w$, which implies that $q(w) = \text{Normal}(w \mid \mu_w, \Sigma_w)$

$$\propto -\frac{E[\lambda]}{2}\sum_{i=1}^{N}\underbrace{(y_i - x_i^T w)^2}_{} - \underbrace{\frac{1}{2}\sum_{k=1}^{d}E[\alpha_k] w_k^2}_{}$$

focus on terms containing $w$: $y_i^2 - 2y_i x_i^T w + \underbrace{(x_i^T w)^2}_{w^T x_i x_i^T w}$ $\qquad w^T \text{diag}(E[\alpha_1], \cdots, E[\alpha_d]) w$.

→ quadratic terms : $-\frac{1}{2}w^T \underbrace{\left(E[\lambda]\sum_{i=1}^{N} x_i x^T + \text{diag}(E[\alpha_1], \cdots, E[\alpha_d])\right)}_{\Sigma_w^{-1}} w$

$\Rightarrow -\frac{1}{2}w^T \Sigma^{-1} w$

→ linear terms : $-\frac{E[\lambda]}{2}(-2y_i x_i^T w) = \underbrace{\left(E[\lambda]\sum_{i=1}^{N} y_i x_i^T\right)}_{\mu_w^T \Sigma_w^{-1}} w.$

↳ $\mu_w^T = \Sigma_w\left(E[\lambda]\sum_{i=1}^{N} y_i x_i^T\right)$

Note

for multivariate Normal dist. $N(\mu, \Sigma)$

PDF: $(2\pi)^{-\frac{k}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(\underbrace{-\frac{1}{2}(w-\mu)^T\Sigma^{-1}(w-\mu)}_{}\right)$

quadratic: $\underbrace{-\frac{1}{2}w^T\Sigma^{-1}w}_{\text{quadratic term}} + \underbrace{\mu^T\Sigma^{-1}w}_{\text{linear term.}} - \frac{1}{2}\mu^T\Sigma^{-1}\mu$

$q(w) = \text{Normal}(w \mid \mu_w, \Sigma_w)$

$$\Sigma_w^{-1} = E[\lambda]\sum_{i=1}^{N} x_i x^T + \text{diag}(E[\alpha_1], \cdots, E[\alpha_d])$$

$$\mu_w = \Sigma_w\left(E[\lambda]\sum_{i=1}^{N} y_i x_i^T\right)$$

$\ln q(\alpha_k)$ : focusing on terms containing $\alpha_k$

$$\frac{N}{2}\ln\lambda - \frac{\lambda}{2}\sum_{i=1}^{N}(y_i - x_i^T w)^2 + \frac{1}{2}\sum_{k=1}^{d}\ln\alpha_k - \frac{1}{2}\sum_{k=1}^{d}\alpha_k w_k^2 + (a_0-1)\sum_{k=1}^{d}\ln\alpha_k - b_0\sum_{k=1}^{d}\alpha_k + (e_0-1)\ln\lambda - f_0\lambda$$

$$\ln q(\alpha_k) \propto E_{\cancel{q(\lambda)},\, q(w)}\left[\frac{1}{2}\ln\alpha_k - \frac{1}{2}\alpha_k w_k^2 + (a_0-1)\ln\alpha_k - b_0\alpha_k\right]$$

no $\lambda$

$$\propto \underbrace{\left(a_0+\frac{1}{2}-1\right)\ln\alpha_k - \left(b_0+\frac{1}{2}E[w_k^2]\right)\alpha_k}_{\text{log density form of a Gamma distribution}}$$

$$q(\alpha_k) = \text{Gamma}(\alpha_k \mid a_k, b_k)$$

$$a_k = a_0 + \frac{1}{2}, \quad b_k = b_0 + \frac{1}{2}E[w_k^2]$$

for $w_k = \text{Normal}(w_k \mid \mu_{w,k}, \Sigma_{w,k})$

$\ln q(\lambda)$ : focusing on terms containing $\lambda$

$$\frac{N}{2}\ln\lambda - \frac{\lambda}{2}\sum_{i=1}^{N}(y_i - x_i^T w)^2 + \frac{1}{2}\sum_{k=1}^{d}\ln\alpha_k - \frac{1}{2}\sum_{k=1}^{d}\alpha_k w_k^2 + (a_0-1)\sum_{k=1}^{d}\ln\alpha_k - b_0\sum_{k=1}^{d}\alpha_k + (e_0-1)\ln\lambda - f_0\lambda$$

$$\ln q(\lambda) \propto E_{\cancel{q(\alpha)},\, q(w)}\left[\frac{N}{2}\ln\lambda - \frac{\lambda}{2}\sum_{i=1}^{N}(y_i - x_i^T w)^2 + (e_0-1)\ln\lambda - f_0\lambda\right]$$

no $\alpha$

$$\underbrace{\left(\frac{N}{2}+e_0-1\right)\ln\lambda - \left(f_0 + \frac{1}{2}\sum_{i=1}^{N}E\left[(y_i - x_i^T w)^2\right]\right)\lambda}_{\text{log density form of a Gamma distribution}}$$

$$\frac{1}{2}\sum_{i=1}^{N}E_{q(w)}\left[y_i^2 - 2y_i x_i^T w + (x_i^T w)^2\right]$$

$$= \frac{1}{2}\sum_{i=1}^{N}\left(y_i^2 - 2y_i x_i^T E[w] + \boxed{E[w^T(x_i^T x)w]}\right)$$ apply Trace trick

$$= \frac{1}{2}\sum_{i=1}^{N}\left(y_i^2 - 2y_i x_i^T \mu_w + \text{Tr}(x_i^T x \cdot E[ww^T])\right)$$

$$= \frac{1}{2}\sum_{i=1}^{N}\left((y_i - x_i^T\mu_w)^2 + \text{Tr}(x_i^T x \cdot E[ww^T])\right)$$

$$q(\lambda) = \text{Gamma}(\lambda \mid e_N, f_N)$$

$$e_N = e_0 + \frac{N}{2}, \quad f_N = f_0 + \frac{1}{2}\sum_{i=1}^{N}\left((y_i - x_i^T\mu_w)^2 + \text{Tr}(x_i^T x \cdot E[ww^T])\right)$$

(b) VI algorithm summary pseudocode

Inputs: data and definitions $q(w) = \text{Normal}(w \mid \mu', \Sigma')$,

$q(\alpha_k) = \text{Gamma}(\alpha_k \mid a', b')$,

$q(\lambda) = \text{Gamma}(\lambda \mid e', f')$

Output: values for $\mu', \Sigma', a', b', e', f'$

1. Initialize $a_0, b_0, e_0, f_0$ in some way.
2. For iteration $t = 1, \cdots, T$
   - update $q(w)$ by setting

$$\Sigma_t^{-1} = E[\lambda] \sum_{i=1}^{N} x_i x^T + \text{diag}(E[\alpha_1], \cdots, E[\alpha_d])$$

$$\mu_t = \Sigma_t \left( E[\lambda] \sum_{i=1}^{N} y_i x_i^T \right)$$

   - update $q(\alpha_k)$ by setting

$$a'_{k,t} = a_0 + \frac{1}{2},$$

$$b'_{k,t} = b_0 + \frac{1}{2} E[w_k^2]$$

   - update $q(\lambda)$ by setting

$$e'_t = e_0 + \frac{N}{2},$$

$$f'_t = f_0 + \frac{1}{2} \sum_{i=1}^{N} \left( (y_i - x_i^T \mu_t)^2 + \text{Tr}(x_i^T x \cdot E[ww^T]) \right)$$

   - Evaluate $\mathcal{L}(\mu', \Sigma', a', b', e', f')$ to assess convergence.

(C)

Variational objective function

$$\mathcal{L} = \underbrace{E_q[\ln p(y, w, \alpha, \lambda)]}_{(a)} - \underbrace{E_q[\ln q(w, \alpha, \lambda)]}_{(b)}$$

(a) $$E_q[\ln p(y,w,\alpha,\lambda)] = \underbrace{E_q[\ln p(y|w,\lambda)]}_{(i)} + \underbrace{E_q[\ln p(w|\alpha)]}_{(ii)} + \underbrace{\sum_{k=1}^{d} E_q[\ln p(\alpha_k)]}_{(iii)} + \underbrace{E_q[\ln p(\lambda)]}_{(iv)}$$

$y_i \overset{ind}{\sim} Normal(x_i^T w, \lambda^{-1})$

$$\begin{aligned}
(i) &= E_q\Big[\sum_{i=1}^{N} \ln\big((2\pi\lambda^{-1})^{-\frac{1}{2}} \exp(-(y_i - x_i^T w)^2/2\lambda^{-1})\big)\Big] \\
&= E_q\Big[\sum_{i=1}^{N}\Big(-\frac{1}{2}\ln\Big(\frac{2\pi}{\lambda}\Big) - \frac{\lambda}{2}(y_i - x_i^T w)^2\Big)\Big] \\
&= E_q\Big[\sum_{i=1}^{N}\Big(-\frac{1}{2}\ln(2\pi) + \frac{1}{2}\ln\lambda - \frac{\lambda}{2}(y_i - x_i^T w)^2\Big)\Big] \\
&= E_q\Big[-\frac{N}{2}\ln(2\pi) + \frac{N}{2}\ln\lambda - \frac{\lambda}{2}\sum_{i=1}^{N}(y_i - x_i^T w)^2\Big] \\
&= -\frac{N}{2}\ln(2\pi) + \frac{N}{2}\ln E_q[\lambda] - \frac{E_q[\lambda]}{2}\sum_{i=1}^{N} E_q[(y_i - x_i^T w)^2] \\
&= -\frac{N}{2}\ln(2\pi) + \frac{N}{2}\ln E_q[\lambda] - \frac{E_q[\lambda]}{2}\sum_{i=1}^{N}\big((y_i - x_i^T \mu_w)^2 + Tr(x_i^T x \cdot E[ww^T])\big)
\end{aligned}$$

$E_q[\lambda] = \frac{e_N}{f_N}$

$E_q[\ln\lambda] = \psi(e_N) - \ln f_N$

$w \sim Normal(0, diag(\alpha_1, \cdots, \alpha_d)^{-1})$

$$(ii) = E_q\Big[\ln\Big[(2\pi)^{-\frac{d}{2}} |\Lambda|^{\frac{1}{2}} \exp\Big(-\frac{1}{2}\underbrace{w^T diag(\alpha_1, \cdots, \alpha_d) w}_{\sum_{k=1}^{d}\alpha_k w_k^2}\Big)\Big]\Big]$$

$\Lambda = diag(\alpha_1, \cdots, \alpha_d)$

$|\Lambda| = \prod_{k=1}^{d}\alpha_k$

$$\begin{aligned}
&= E_q\Big[-\frac{d}{2}\ln(2\pi) + \frac{1}{2}\sum_{k=1}^{d}\ln\alpha_k - \frac{1}{2}\sum_{k=1}^{d}\alpha_k w_k^2\Big] \\
&= -\frac{d}{2}\ln(2\pi) + \frac{1}{2}\sum_{k=1}^{d} E_q[\ln\alpha_k] - \frac{1}{2}\sum_{k=1}^{d} E_q[\alpha_k]\, E_q[ww^T]
\end{aligned}$$

$E_q[w] = \mu_w$

$E_q[ww^T] = \mu_w^2 + \Sigma_w$

$E[w_k^2] = diag(E_q[ww^T])$

$E_q[\alpha_k] = \frac{a_k}{b_k}$

$E_q[\ln\alpha_k] = \psi(a_k) - \ln b_k$

$\alpha_k \overset{iid}{\sim} Gamma(a_0, b_0)$

$$\begin{aligned}
(iii) &= \sum_{k=1}^{d}\big(a_0\ln b_0 - \ln\Gamma(a_0) + (a_0 - 1)E_q[\ln\alpha_k] - b_0 E_q[\alpha_k]\big) \\
&= d(a_0\ln b_0 - \ln\Gamma(a_0)) + (a_0 - 1)\sum_{k=1}^{d} E_q[\ln\alpha_k] - b_0 \sum E_q[\alpha_k]
\end{aligned}$$

$\lambda \sim \text{Gamma}(e_0, f_0)$

$$(iv) = e_0 \ln f_0 - \ln\Gamma(e_0) + (e_0 - 1)E_q[\ln\lambda] - f_0 E_q[\lambda]$$

$$E_q[\lambda] = \frac{e_N}{f_N}$$
$$E_q[\ln\lambda] = \Psi(e_N) - \ln f_N$$

$$(b) = -E_q[\ln q(w, \alpha, \lambda)]$$ the entropy $H(\,)$

total entropy = $\Sigma$ individual entropies

$$= H(q(w)) + \sum_{k=1}^{d} H(q(\alpha_k)) + H(q(\lambda))$$

Entropy of multivariate normal
$$H(q(w)) = \frac{d}{2}\ln(2\pi e) + \frac{1}{2}\ln|\Sigma_w|$$
$$= \frac{d}{2}(\ln(2\pi)+1) + \frac{1}{2}\ln|\Sigma_w|$$

Entropy of Gamma
$$\sum_{k=1}^{d} H(q(\alpha_k)) = \sum_{k=1}^{d}\left(a_k - \ln b_k + \ln\Gamma(a_k) + (1-a_k)\Psi(a_k)\right)$$

$a_k$ is constant: $a_k = a_0 + \frac{1}{2}$

$$= d\left(a_k + \ln\Gamma(a_k) + (1-a_k)\Psi(a_k)\right) - \sum_{k=1}^{d}\ln b_k$$

Entropy of Gamma
$$H(q(\lambda)) = \left(e_N - \ln f_N + \ln\Gamma(e_N) + (1-e_N)\Psi(e_N)\right)$$

$\therefore$ Variational objective function

$$\mathcal{L}_t(\mu', \Sigma', a', b', e', f')$$

$$\left.\begin{aligned}
= &-\frac{N}{2}\ln(2\pi) + \frac{N}{2}\ln E_q[\lambda] - \frac{E_q[\lambda]}{2}\sum_{i=1}^{N}\left((y_i - x_i^T\mu_w)^2 + Tr(x_i^T x \cdot E[ww^T])\right) \\
&-\frac{d}{2}\ln(2\pi) + \frac{1}{2}\sum_{k=1}^{d}E_q[\ln\alpha_k] - \frac{1}{2}\sum_{k=1}^{d}E_q[\alpha_k]E_q[w_k^2] \\
&+ d(a_0\ln b_0 - \ln\Gamma(a_0)) + (a_0-1)\sum_{k=1}^{d}E_q[\ln\alpha_k] - b_0\Sigma E_q[\alpha_k] \\
&+ e_0\ln f_0 - \ln\Gamma(e_0) + (e_0-1)E_q[\ln\lambda] - f_0 E_q[\lambda]
\end{aligned}\right\} E_q[\ln p]$$

$$\left.\begin{aligned}
&+ \frac{d}{2}(\ln(2\pi)+1) + \frac{1}{2}\ln|\Sigma_w| \\
&+ d\left(a_t' + \ln\Gamma(a_t') + (1-a_t')\Psi(a_t')\right) - \sum_{k=1}^{d}\ln b_{k,t}' \\
&+ e_t' - \ln f_t' + \ln\Gamma(e_t') + (1-e_t')\Psi(e_t')
\end{aligned}\right\} \text{entropies}$$

(b) VI algorithm summary pseudocode (how it is translated in actual code)

Inputs: data and definitions $q(w) = \text{Normal}(w \mid \mu', \Sigma')$,

$q(\alpha_k) = \text{Gamma}(\alpha_k \mid a', b')$,

$q(\lambda) = \text{Gamma}(\lambda \mid e', f')$

Output: values for $\mu', \Sigma', a', b', e', f'$

1. Initialize $a_0, b_0, e_0, f_0$ in some way.
2. For iteration $t = 1, \cdots, T$

① update $q(w)$ by setting

XTX

precision_w $\Sigma_t^{-1} = E[\lambda] \sum_{i=1}^{N} x_i x_i^T + \text{diag}(E[\alpha_1], \cdots, E[\alpha_d])$

E_lambda np.diag(E_alpha)

np.linalg.inv $\mu_t = \Sigma_t \left( E[\lambda] \sum_{i=1}^{N} y_i x_i^T \right)$

Sigma_w XTy

mu_X $E_q[w] = \mu_w$

E_wwT $E_q[ww^T] = \mu_w^2 + \Sigma_w$

E_wk2 $E[w_k^2] = \text{diag}(E_q[ww^T])$

② update $q(\alpha_k)$ by setting

$a'_{k,t} = a_0 + \frac{1}{2}$, constant

$b'_{k,t} = b_0 + \frac{1}{2} E[w_k^2]$

E_alpha $E_q[\alpha_k] = \frac{a_k}{b_k}$

E_log_alpha $E_q[\ln \alpha_k] = \psi(a_k) - \ln b_k$

③ update $q(\lambda)$ by setting

E_lambda $E_q[\lambda] = \frac{e_N}{f_N}$

E_log_lambda $E_q[\ln \lambda] = \psi(e_N) - \ln f_N$

$e'_t = e_0 + \frac{N}{2}$, constant.

$f'_t = f_0 + \frac{1}{2} \sum_{i=1}^{N} \left( (y_i - x_i^T \mu_t)^2 + Tr(x_i^T x \cdot E[ww^T]) \right)$

$\sum_{i=1}^{N} \left( y_i^2 - 2 y_i x_i^T \mu_w + Tr(x_i^T x \cdot E[ww^T]) \right)$

sum_sq_err

④ Evaluate $\mathcal{L}(\mu', \Sigma', a', b', e', f')$ to assess convergence

next page

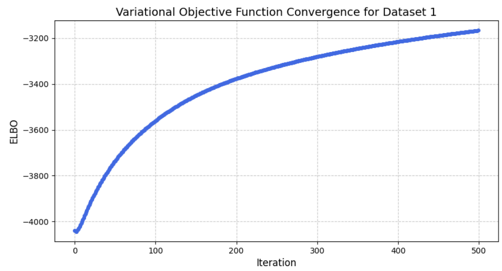
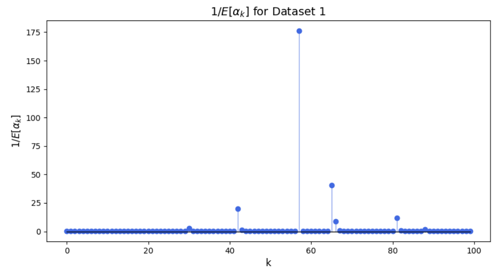
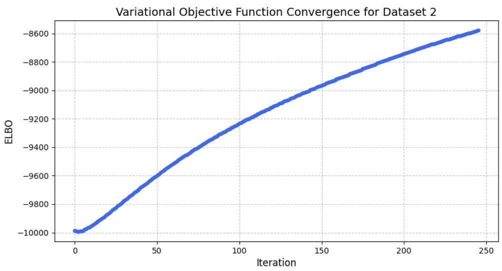
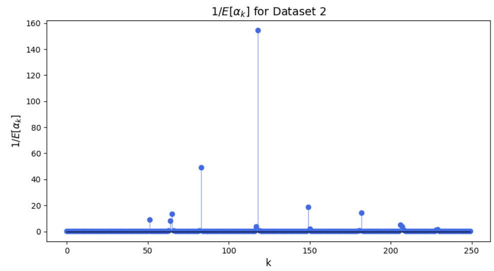
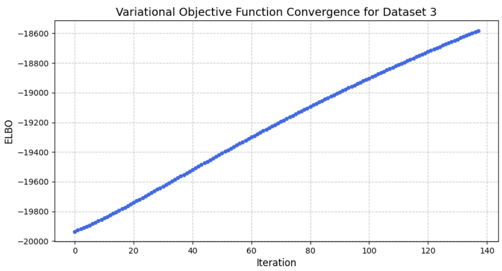
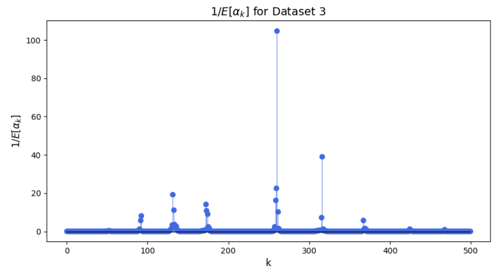
④ Evaluate $\mathcal{L}(\mu', \Sigma', a', b', e', f')$ to assess convergence

$$\mathcal{L} = \underbrace{E_q[\ln p(y, w, \alpha, \lambda)]}_{\text{part (a)}} - \underbrace{E_q[\ln q(w, \alpha, \lambda)]}_{\text{part (b)}}$$

(a) $$E_q[\ln p(y, w, \alpha, \lambda)] = \underbrace{E_q[\ln p(y|w,\lambda)]}_{\substack{\text{(i)} \\ \text{log\_p\_y}}} + \underbrace{E_q[\ln p(w|\alpha)]}_{\substack{\text{(ii)} \\ \text{log\_p\_w}}} + \underbrace{\sum_{k=1}^{d} E_q[\ln p(\alpha_k)]}_{\substack{\text{(iii)} \\ \text{log\_p\_alpha}}} + \underbrace{E_q[\ln p(\lambda)]}_{\substack{\text{(iv)} \\ \text{log\_p\_lambda}}}$$

(b) $$-E_q[\ln q(w, \alpha, \lambda)] = \underbrace{H(q(w))}_{\text{h\_w}} + \underbrace{\sum_{k=1}^{d} H(q(\alpha_k))}_{\text{h\_alpha}} + \underbrace{H(q(\lambda))}_{\text{h\_lambda}}$$

**Problem 2**

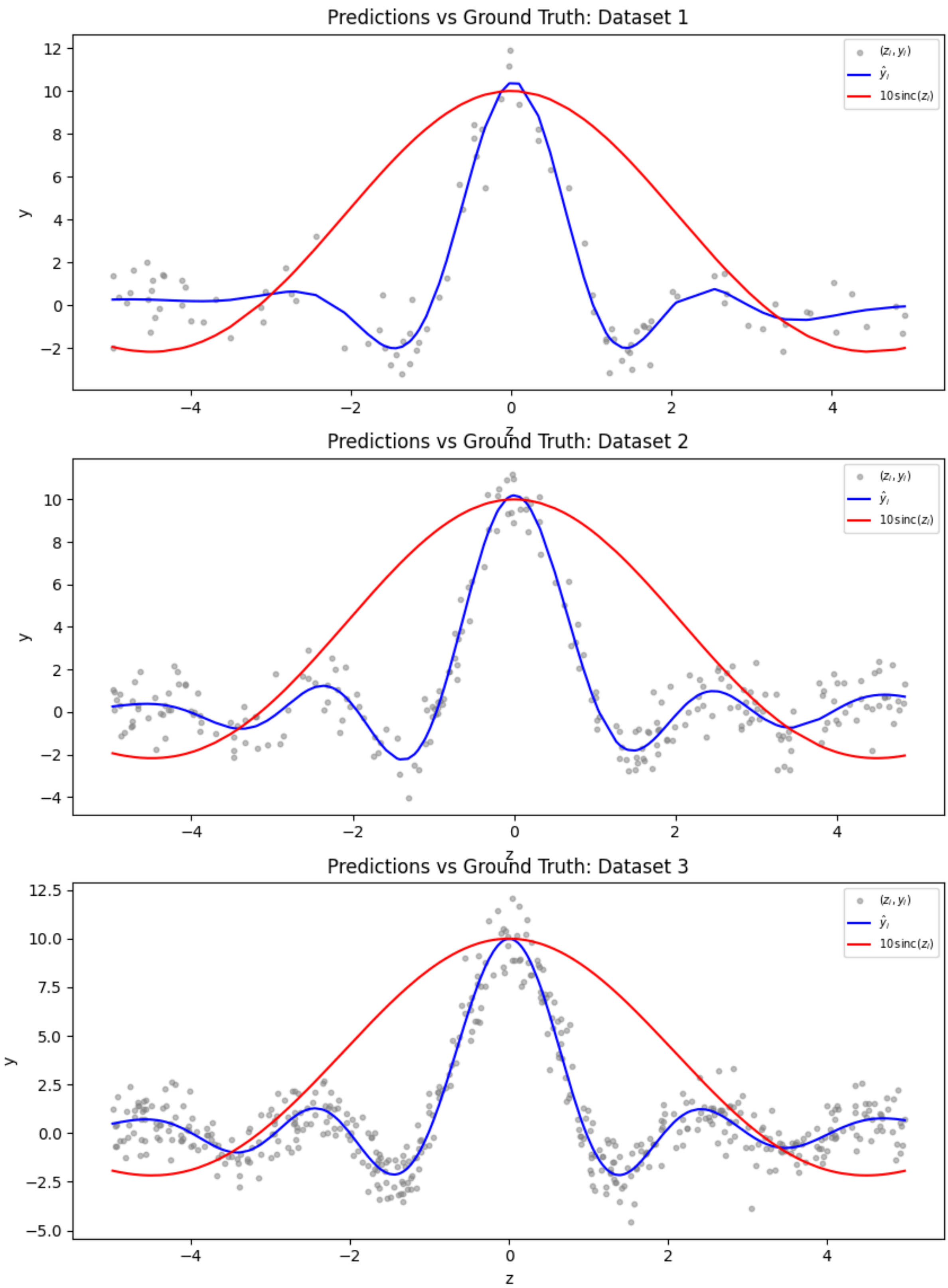
| (a) plot variational objective function | (b) plot $1/\mathrm{E}[\alpha_k]$ |
|---|---|
|  |  |
|  |  |
|  |  |



(c) $1/E[\lambda]$ at final iteration
Dataset 1: 0.926
Dataset 2: 1.112
Dataset 3: 1.022

Plots for (d) is on the next page

(d)


Predictions vs Ground Truth: Dataset 1
Predictions vs Ground Truth: Dataset 2
Predictions vs Ground Truth: Dataset 3
$(z_i, y_i)$
$\hat{y}_i$
$10\,\mathrm{sinc}(z_i)$
y
z

# Python Code for Problem 2

```python
import numpy as np
from scipy.special import digamma, gammaln
import pandas as pd
import matplotlib.pyplot as plt
import os

def read_files(folder):
    X_set1 = pd.read_csv(os.path.join(folder, 'X_set1.csv'), header=None).values
    X_set2 = pd.read_csv(os.path.join(folder, 'X_set2.csv'), header=None).values
    X_set3 = pd.read_csv(os.path.join(folder, 'X_set3.csv'), header=None).values

    y_set1 = pd.read_csv(os.path.join(folder, 'y_set1.csv'), header=None).values.flatten()
    y_set2 = pd.read_csv(os.path.join(folder, 'y_set2.csv'), header=None).values.flatten()
    y_set3 = pd.read_csv(os.path.join(folder, 'y_set3.csv'), header=None).values.flatten()

    z_set1 = pd.read_csv(os.path.join(folder, 'z_set1.csv'), header=None).values.flatten()
    z_set2 = pd.read_csv(os.path.join(folder, 'z_set2.csv'), header=None).values.flatten()
    z_set3 = pd.read_csv(os.path.join(folder, 'z_set3.csv'), header=None).values.flatten()
    return X_set1, X_set2, X_set3, y_set1, y_set2, y_set3, z_set1, z_set2, z_set3

def run_vi(X, y, a0=1e-16, b0=1e-16, e0=1, f0=1, max_iter=500, tol=1e-6):
    N, d = X.shape

    # 1. Initialize expectations
    E_alpha = np.full(d, a0 / b0)
    E_lambda = e0 / f0

    # Precompute XTX and XTy for speed
    XTX = X.T @ X
    XTy = X.T @ y
    # Precompute constant part of a_k and e_N
    ak = a0 + 0.5
    eN = e0 + N / 2.0

    # Storage for tracking ELBO
    elbo_history = []

    for i in range(max_iter):
        # --- STEP 1: Update q(w) ---
        # Precision matrix (Sigma inverse)
        precision_w = E_lambda * XTX + np.diag(E_alpha)
        Sigma_w = np.linalg.inv(precision_w)
        mu_w = Sigma_w @ (E_lambda * XTy)

        # Expectations needed for other updates
        E_wwT = Sigma_w + np.outer(mu_w, mu_w)
        E_wk2 = np.diag(E_wwT)

        # --- STEP 2: Update q(alpha_k) ---
        # ak is constant, only bk changes
        bk = b0 + 0.5 * E_wk2
        E_alpha = ak / bk
        E_log_alpha = digamma(ak) - np.log(bk)

        # --- STEP 3: Update q(lambda) ---
        # Note: E[(y - Xw)^2] = yTy - 2yX*mu_w + Tr(XTX * E[wwT])
        yTy = np.dot(y, y)
        sum_sq_err = yTy - 2 * np.dot(mu_w, XTy) + np.trace(XTX @ E_wwT)
        # eN is constant, only fN changes
        fN = f0 + 0.5 * sum_sq_err
        E_lambda = eN / fN
        E_log_lambda = digamma(eN) - np.log(fN)

        # --- STEP 4: Calculate exact variational objective function ---
        # part (a): Joint expectations
```

```python
        log_p_y = -(N/2)*np.log(2*np.pi) + (N/2)*E_log_lambda - 0.5*E_lambda*sum_sq_err
        log_p_w = -(d/2)*np.log(2*np.pi) + 0.5*np.sum(E_log_alpha) - 0.5*np.sum(E_alpha * E_wk2)
        log_p_alpha = d*(a0*np.log(b0)-gammaln(a0)) + (a0-1)*np.sum((a0 - 1) * E_log_alpha - b0 * E_alpha)
        log_p_lambda = e0*np.log(f0) - gammaln(e0) + (e0-1)*E_log_lambda - f0*E_lambda
        log_joint = log_p_y + log_p_w + log_p_alpha + log_p_lambda

        # part (b): Entropies
        h_w = (d/2)*np.log(2*np.pi + 1) + 0.5*np.log(np.linalg.det(Sigma_w))
        h_alpha = d*(ak + gammaln(ak) + (1-ak)*digamma(ak)) - np.sum(np.log(bk))
        #h_alpha = np.sum(ak - np.log(bk) + gammaln(ak) + (1-ak)*digamma(ak))
        h_lambda = eN - np.log(fN) + gammaln(eN) + (1-eN)*digamma(eN)
        entropies = h_w + h_alpha + h_lambda

        current_elbo = log_joint + entropies
        elbo_history.append(current_elbo)

    return mu_w, Sigma_w, E_alpha, E_lambda, elbo_history


# 1. Run your inference
def plot_variational_objective(elbo_history, dataset_name):
    plt.figure(figsize=(10, 5))
    plt.plot(elbo_history, color='royalblue', linewidth=2, marker='o', markersize=4)
    plt.title(f'Variational Objective Function Convergence for {dataset_name}', fontsize=14)
    plt.xlabel('Iteration', fontsize=12)
    plt.ylabel('ELBO', fontsize=12)
    plt.grid(True, linestyle='--', alpha=0.7)
    plt.show()


def plot_inv_e_alpha(E_alpha, dataset_name):
    inv_E_alpha = 1.0 / np.maximum(E_alpha, 1e-15)
    plt.figure(figsize=(10, 5))
    markerline, stemlines, baseline = plt.stem(range(len(inv_E_alpha)), inv_E_alpha)
    plt.setp(markerline, marker='o', color='royalblue', markersize=6)
    plt.setp(stemlines, color='royalblue', linewidth=1, alpha=0.6)
    plt.setp(baseline, color='black', linewidth=1) # This styles the stem's baseline
    plt.title(f"$1/E[\\alpha_k]$ for {dataset_name}", fontsize=14)
    plt.xlabel("k", fontsize=12)
    plt.ylabel("$1/E[\\alpha_k]$", fontsize=12)
    plt.show()


folder = '/Users/jihyunpark/Documents/Columbia/2026 Spring/Bayesian/HW2/data_csv'
X_set1, X_set2, X_set3, y_set1, y_set2, y_set3, z_set1, z_set2, z_set3 = read_files(folder)

# (a)
mu_w1, Sigma_w1, E_alpha1, E_lambda1, elbo_history1 = run_vi(X_set1, y_set1)
plot_variational_objective(elbo_history1, "Dataset 1")
mu_w2, Sigma_w2, E_alpha2, E_lambda2, elbo_history2 = run_vi(X_set2, y_set2)
plot_variational_objective(elbo_history2, "Dataset 2")
mu_w3, Sigma_w3, E_alpha3, E_lambda3, elbo_history3 = run_vi(X_set3, y_set3)
plot_variational_objective(elbo_history3, "Dataset 3")

#(b)
plot_inv_e_alpha(E_alpha1, "Dataset 1")
plot_inv_e_alpha(E_alpha2, "Dataset 2")
plot_inv_e_alpha(E_alpha3, "Dataset 3")

#(c)
print(f"Dataset 1: {E_lambda1:.3f}")
print(f"Dataset 2: {E_lambda2:.3f}")
print(f"Dataset 3: {E_lambda3:.3f}")


#(d) Predictions and comparison to ground truth
def plot_zi_yi(ax, X, y, z, mu_w, dataset_name):
    # (d) Predictions: sort by z for clean line plot
    y_hat = X @ mu_w
```

```python
    sort_idx = np.argsort(z)
    z_s, y_hat_s = z[sort_idx], y_hat[sort_idx]
    sinc_s = 10 * np.sinc(z_s / np.pi)  # ground truth: 10*sin(z)/z

    ax.scatter(z, y, s=10, alpha=0.5, color="gray", label=r"$(z_i, y_i)$")
    ax.plot(z_s, y_hat_s, color="blue", label=r"$\hat{y}_i$")
    ax.plot(z_s, sinc_s, color="red", label=r"$10\,\mathrm{sinc}(z_i)$")
    ax.set_xlabel("z")
    ax.set_ylabel("y")
    ax.set_title("Predictions vs Ground Truth: " + dataset_name)
    ax.legend(fontsize=7)

fig, axs = plt.subplots(3, 1, figsize=(10, 15))
plot_zi_yi(axs[0], X_set1, y_set1, z_set1, mu_w1, "Dataset 1")
plot_zi_yi(axs[1], X_set2, y_set2, z_set2, mu_w2, "Dataset 2")
plot_zi_yi(axs[2], X_set3, y_set3, z_set3, mu_w3, "Dataset 3")
plt.tight_layout()
plt.show()
```